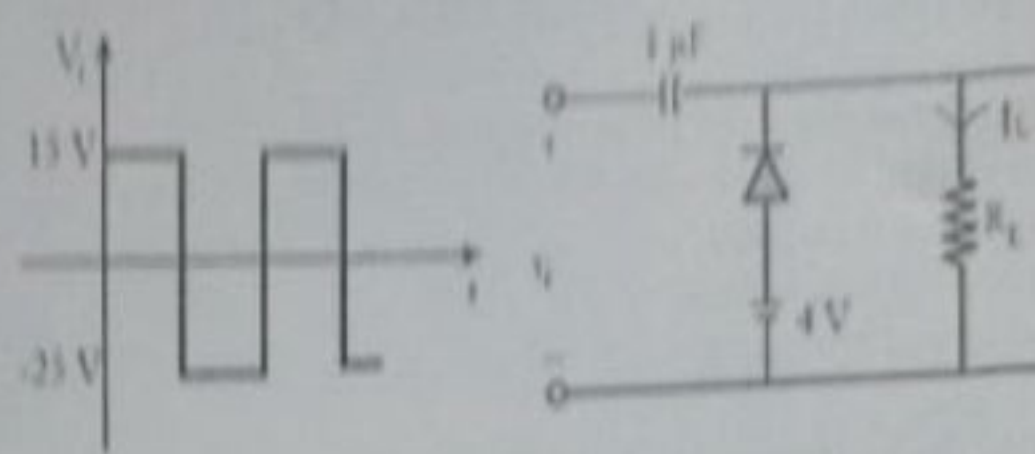


Q1. Select the correct answer for the following questions:

1. The addition of pentavalent impurity to a semiconductor creates many
a. Holes. b. Free electrons. c. Bound electrons. d. Valence electrons.
2. The battery connections required to forward bias a pn junction are
a. -ve terminal to p and +ve terminal to n
b. +ve terminal to p and -ve terminal to n
c. -ve terminal to p and -ve terminal to n
d. None of the above
3. In a capacitor-input filter that results in ripple voltage, the increasing of the load resistance affects the DC voltage.
a. Increasing. b. Decreasing. c. Does not affect.
4. In diode configuration circuit, the average resistance is defined as
a. A straight line between limits of operation.
b. A point on the characteristics.
c. A tangent line at the Q-point.
d. None of the above.
5. An n -type semiconductor is
a. Positively charged. b. Negatively charged. c. Electrically neutral d. None of the above.
6. The required energy E_g for each semiconductor electron in the valence band to move to the conduction band is ordered from high to low as.....
a. $Ge < GaAs < Si$. b. $Si < Ge < GaAs$. c. $GaAs < Ge < Si$. d. $Ge < Si < GaAs$.
7. For the circuit shown below, $V_1 = 3\text{ V}$ and $V_2 = 1\text{ V}$. If the resistor $R_2 = 1.5\text{ k}\Omega$ and the diode is a simplified model diode, then the current I is equal to.....
a. 0 mA . b. 1.33 mA . c. 1.53 mA . d. 0.47 mA .

Q2. For the given circuit below, if the $R_L = 5\text{ k}\Omega$ and the diode is ideal, answer the following:

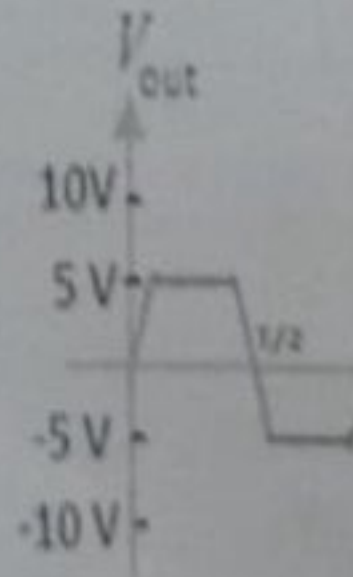
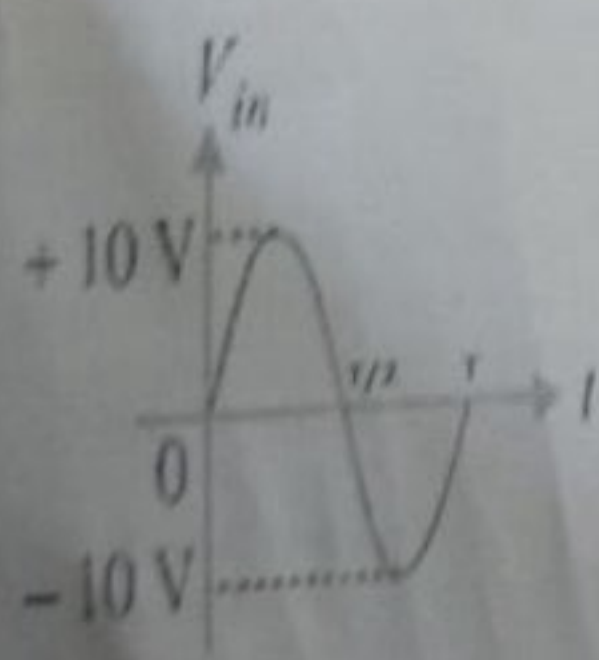


- 2.1 What is the required time constant that should be considered to guarantee that the capacitor is not discharged during the non-conducting (OFF) interval of the diode.
- 2.2. What is the output during the positive half-cycle?
- 2.3 What is the current through the resistor R_L during the positive half cycle?

Q3. For the following input and output waveforms, answer the following:

- 3.1. What is the name of the circuit that leads to the output waveform?
- 3.2. Draw the designed circuit that gives the output waveform based on the input waveform. Assume the diode is ideal.

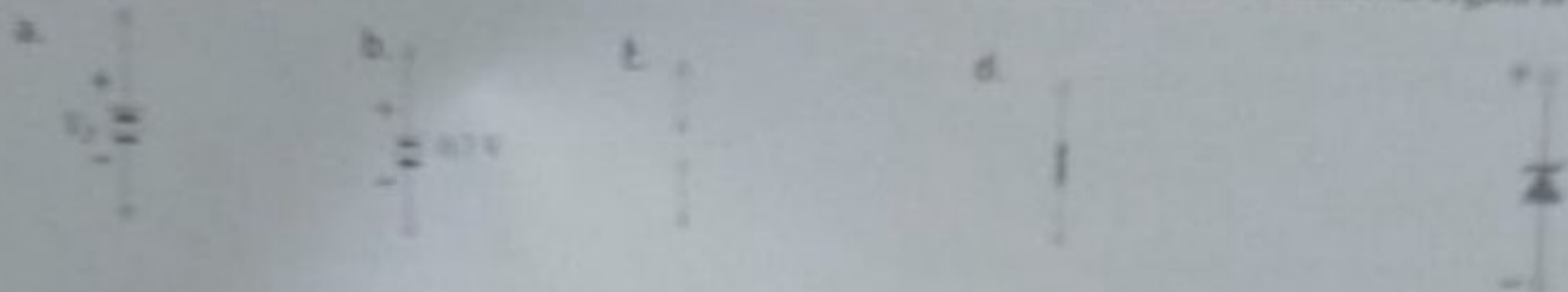
Designed Circuit



8. In a bridge full wave rectifier, the DC voltage, for input voltage V_m and simplified model of diode, is given as.....
- a. $0.636 (V_m - 0.3)$ b. $0.636 (2V_m - 1.6)$ c. $0.636 (V_m - 0.6)$ d. $0.636 (2V_m - 0.3)$

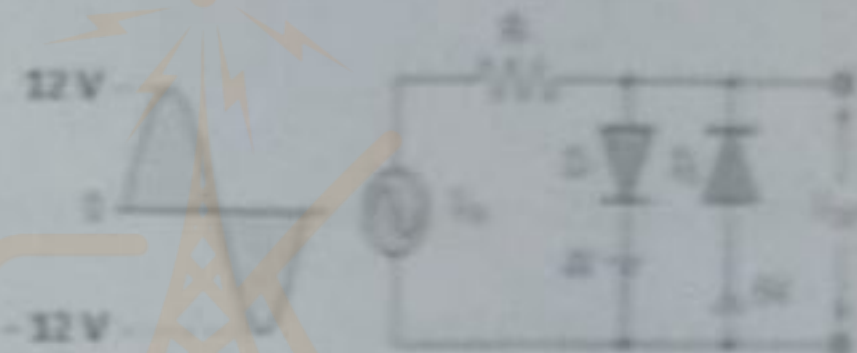
9. A reverse bias pn junction has
- a. Very narrow depletion layer
- b. Almost no current
- c. Very low resistance
- d. Large current flow

10. The best figure that represents the given practical Zener diode in the breakdown region is



11. For the following circuit with assuming that a Silicon simplified model diode for $D1$ and $D2$, the for the negative half cycle is equal to.....

- a. 5.3 V
- b. -8.7 V
- c. 4.3 V
- d. -6.7 V
- e. 3.3 V



12. The following circuit represents a and after a full cycle of input voltage, the between the terminals a and b is..... (The two diodes are ideal)

- a. A half-wave circuit, V_m .
- b. A full-wave circuit, $2 V_m$.
- c. A half-wave circuit, $- V_m$.
- d. A full-wave circuit, $3 V_m$.
- e. A half-wave circuit, $- 3 V_m$.
- f. A full-wave circuit, $- 2 V_m$.

